Fake & Spam Review Detection

Using Machine Learning: RoBERTa + GatedFusion · TextSCNN · XGBoost · Stacking Ensemble

NLP | Deep Learning | Ensemble Methods | 3-Class Classification

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Amazon 45K+ review dataset

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Problem Overview

Fake Reviews

Mislead consumers, inflate/deflate product ratings artificially

Spam Reviews

Promotional/irrelevant content with links and calls-to-action

Why ML?

Manual moderation infeasible at scale — millions of reviews daily

3-Class Task

REAL

Genuine, authentic customer review

FAKE

Deceptive, fabricated review

SPAM

Promotional or irrelevant content

Input: text + rating + category → Output: label

Problem Background - What is Deceptive Spam?

Definition: Fictitious opinions deliberately written to sound authentic. Reviews that are irrelevant, off-topic, promotional, or violate platform guidelines — regardless of whether they're genuine or not.

Real-world impact:

- Businesses hire people to write undeserving positive reviews
- Or unjust negative reviews to damage competitors
- Humans perform poorly: only **57.33%** accuracy [Ref: test of Ott et al.]

Examples of Deceptive Spam

1. **Off-Topic Content (**Complaining about delivery, not the product**)**
   - Review on a product: "Ordered this for my son. Also the delivery guy was absolutely terrible and threw the package at my door. Amazon logistics is a complete disaster and needs to be fixed immediately.“

2. **Promotional / Advertising Content**
   - "Great product! Also check out my YouTube channel where I review products like this!“

3. **Repeated / Duplicate Reviews**
   - Same paragraph copy-pasted across 15 different phone case listings.

4. **Irrelevant External Links (**Very common spam pattern)

Examples of Deceptive Spam

5. Gibberish or Placeholder Text

- "asdfghjkl great product lorem ipsum buy now“

6. **Keyword Stuffing**

- "Best laptop laptop deals cheap laptop buy laptop laptop under 500 laptop gaming laptop“

7. **Very Short Meaningless Reviews ("ok" / "good" / "fine" / "yes“)**

Problem Background - What is a Fake (AI Generated) Review?

- A **fake AI-generated review** is a review written by an AI rather than a real human who actually used the product.
- These reviews are designed to look convincing, but they are not based on real experience

Potential side effects on a shopping site like Amazon:

- **Consumer mistrust**
- **Poor purchase decisions**
- **Damage to brands**
- **Algorithm manipulation**
- **Legal risks** (Platforms can face lawsuits or regulatory scrutiny for letting AI-generated fake reviews run rampant)

Example Cases of Fake Reviews

1. Sentiment-Rating Mismatch: Review text says something negative but gives 5 stars, or vice versa.

- Worst blender I have ever owned. Burns food, leaks, loud as a jet engine. Absolute garbage. Do not buy."
  — Text is negative but rating is 5. Clear manipulation signal.

2. Overly Generic Language:

- "Great product! Very good quality. Highly recommend to everyone. Best purchase ever!“

3. Extreme Sentiment (No Nuance)

- "ABSOLUTELY PERFECT IN EVERY WAY. LIFE CHANGING PRODUCT. 10/10 NO COMPLAINTS"
  — Real users almost always mention at least one minor issue.

Example Cases of Fake Reviews

4. First Person Overuse / Scripted Tone

- "I received this product and I was amazed. I showed it to my family. I use it every day. I love it.“

5**. No Mention of Use Case**

- Real reviewers say *how* and *why* they use a product. Fake ones don't.
  > Real: "I bought this for hiking in rainy weather, the grip is excellent on wet rocks"
  > Fake: "Nice product! Great quality! Good value! Fast shipping! Happy purchase! Highly recommend! Will order more!"

Why Neural Networks? Limitations of Traditional Methods

Traditional approach (Feature Engineering):

- Hand-crafted features: n-grams, POS tags etc
- **Problem:** Cannot capture latent semantic meaning
- Requires domain expertise

Neural Network advantage:

- Learns continuous vector representations automatically
- Captures word order AND sentence order
- Semantic-aware representations

Different sentences in a review have different importance: need **attention mechanism.**

Dataset & Features

45K+	3	17	3
Reviews	Classes	Meta Features	Dataset Splits

Text Features

- Review body text (primary)
- TF-IDF bigrams (XGBoost)
- RoBERTa token embeddings

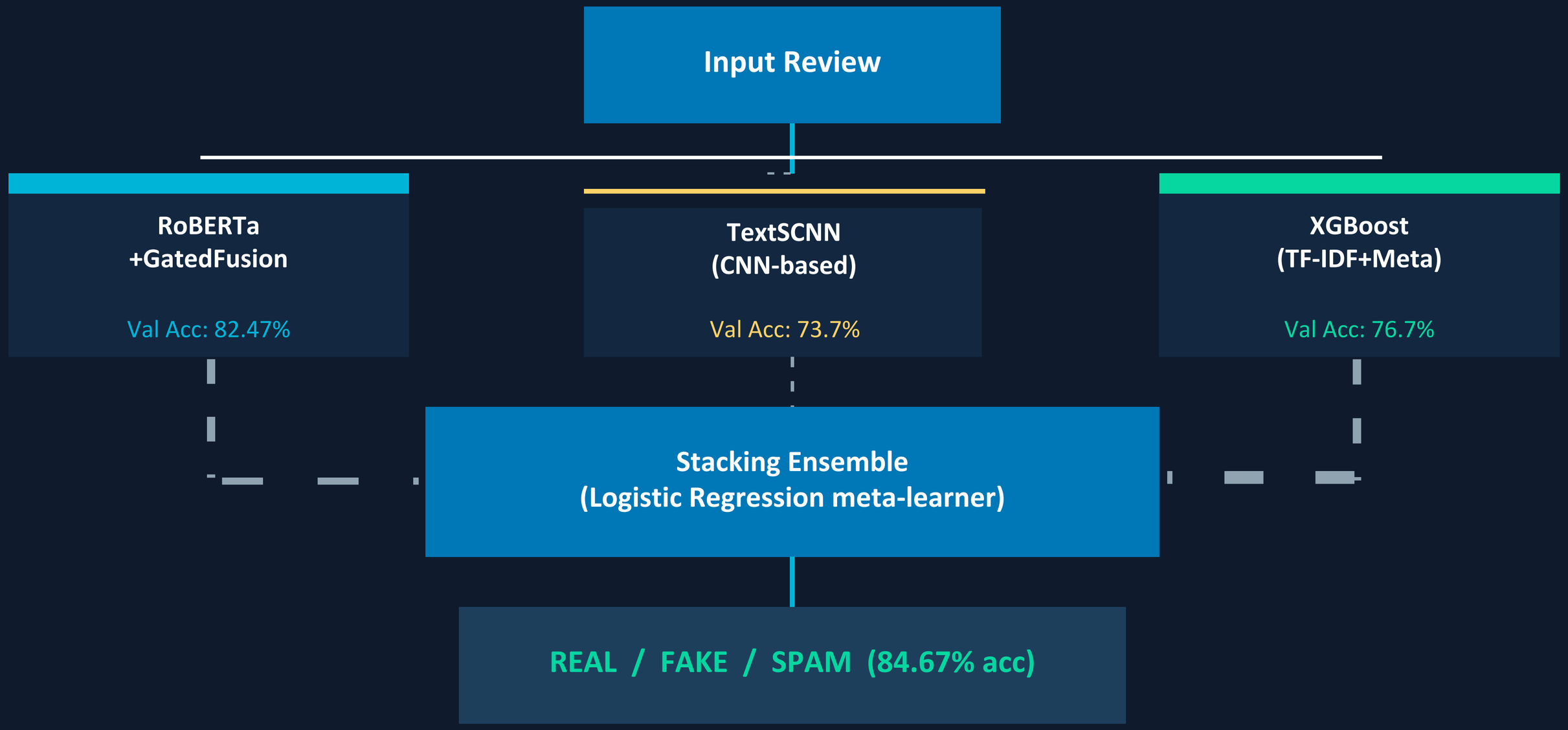
Metadata (16 features)

- Review length, word count
- Exclamation / URL counts
- avg_word_length+punctuation density promotional message
- Rating vs sentiment mismatch
- Sentence caps & all-caps ratio

Data Splits

- Train: ~79.875% of 45K+
- Validation: ~10.125%
- Test: ~10%

System Architecture



Model 1 — RoBERTa + GatedFusion

Review Text

RoBERTa-base
(Transformer, 12 layers)

[CLS] Embedding (768-d)

Gated Fusion + Metadata MLP

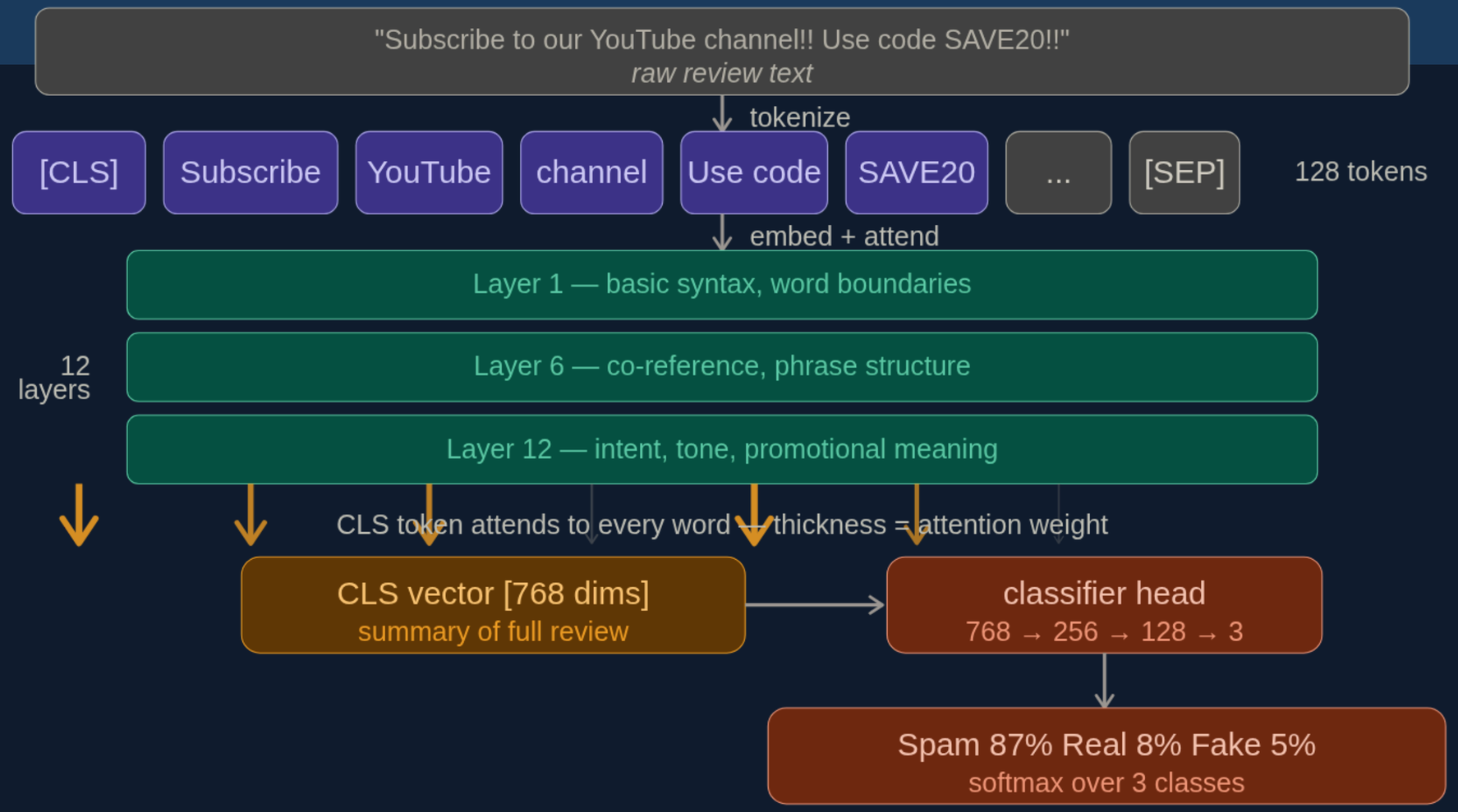
3-class Classifier (Real / Spam / Fake)

Key Design Points

- RoBERTa-base pretrained on large corpora for deep contextual understanding
- GatedFusion layer: learns to weight text vs metadata adaptively
- 16-dim metadata (sentiment, caps, URL count...) fed through 2-layer MLP
- Gate = $\sigma(W \cdot [\text{text}; \text{meta}])$ — suppresses noisy metadata when text is decisive
- Trained with weighted cross-entropy to handle class imbalance
- AdamW optimizer + linear warm-up scheduler over 3 epochs

How RoBERTa reads a review

input → tokens → 12 transformer layers → CLS vector → class



Why RoBERTa understands this review

It was pre-trained on 160 GB of text — it already knows "Subscribe to our channel" means self-promotion. "Use code" means affiliate spam. This is transfer learning. Fine-tuning on your 44k reviews sharpens this for your specific Real/Spam/Fake task.

Where RoBERTa alone cannot decide

These reviews have identical text signals — only metadata separates them

Case 1 — same text, opposite intent

"Great product! Works perfectly.
Love it! Highly recommend."
Rating: ★★★★★ verified: YES

↓

RoBERTa: Real 71%

True label: REAL ✓

identical

"Great product! Works perfectly.
Love it! Highly recommend."
Rating: ★★★★★ verified: NO

↓

RoBERTa: Real 71%

True label: FAKE ✗

Case 2 — text looks negative, rating says 5 stars

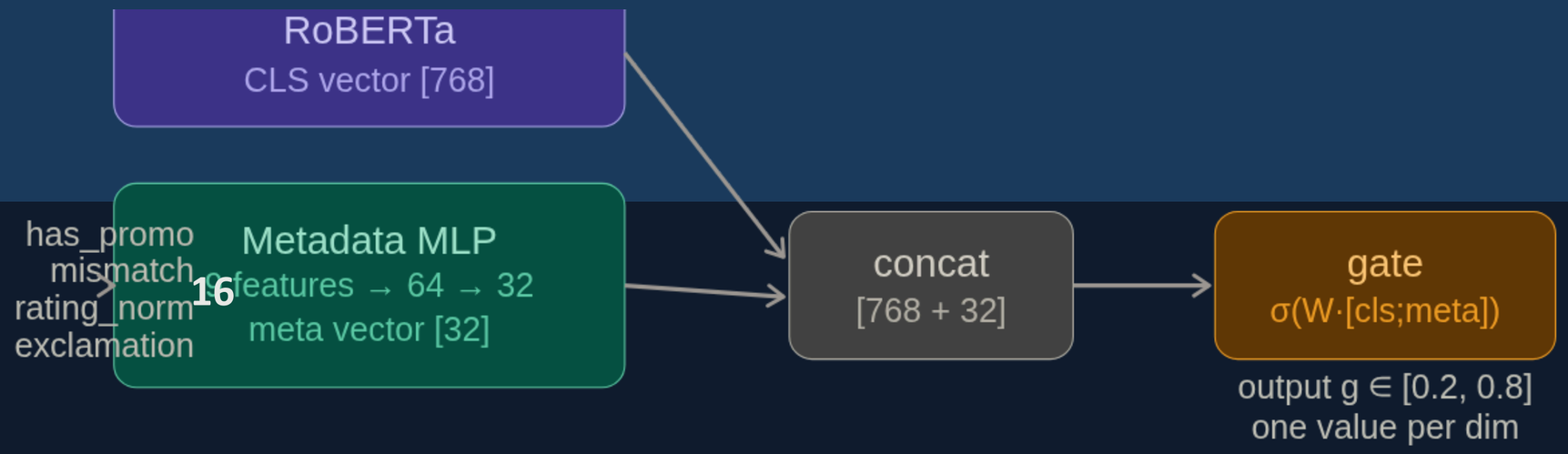
"Terrible. Broke after 2 days. Battery dead. Waste of money."
Rating: ★★★★★ — mismatch signal = 1.7 (very high)

→ RoBERTa alone: Real 55%

(reads negative sentiment → not promotional → hesitat

Solution → gate metadata into the decision

verified=NO and mismatch=1.7 are invisible to RoBERTa's attention mechanism



Fusion formula

fused = g × CLS + (1 − g) × meta_expanded

g close to 1.0 → trust text more g close to 0.2 → trust metadata more

g is learned — different for each of the 768 feature dimensions

Gate value examples

Sarcastic fake review

"Oh wow, broke in 5 mins, AMAZING"

gate g ≈ 0.75 → text dominates

RoBERTa reads sarcasm — metadata can't

Sophisticated fake

"Perfect product! Love it!"

gate g ≈ 0.25 → metadata dominates

Text looks real — mismatch reveals spam

↓

classifier 768 → 256 → 128 → 3

Model 2: TextSWNN · Sentence-Weighted Neural Network

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Document representation and feature combination for deceptive spam review detection

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ABSTRACT

Deceptive spam reviews of products or service are harmful for customers in decision making. Existing approaches to detect deceptive spam reviews are concerned in feature designing. Hand-crafted features can show some linguistic phenomena, however can hardly reveal the latent semantic meaning of the review. We present a neural network based model to learn the representation of reviews. The model makes a hard attention through the composition from sentence representation into document representation. Specifically, we compute the importance weights of each sentence and incorporate them into the composition process of document representation. In the mixed-domain detection experiment, the results verify the effectiveness of our model by comparing with other neural network based methods. As the feature selection is very important in this direction, we make a feature combination to enhance the performance. Then we get 86.1% F1 value which outperform the state-of-the-art method. In the cross-domain detection experiment, our method has better robustness.

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1. Introduction

Deceptive opinion spam detection is an urgent and meaningful task in the field of natural language processing. By continuous growth of the user-generated reviews, the appearance of deceptive opinion spam arouses people's attention [1–4]. Deceptive opinion spam is the review with fictitious opinions which are deliberately written to sound authentic [5]. For commercial motive, some businesses hire people to write undeserving positive reviews to promote the objects or unjust negative reviews to damage the reputations of the objects [6]. It is very difficult distinguish

of neural network based models in the natural language processing tasks currently, the document-level representation can be learnt by neural network based models, and be used as features of the review.

In this work, we try to make a comparison and analysis between representation learning algorithms and conventional features while solving the problem. We present a novel method which is sentence weighted neural network (SWNN) model to learn the document-level representation of the review and detect spam reviews. Learning the representation of the document can capture

https://www.researchgate.net/publication/314200871_Document_Representation_and_Feature_Combination_for_Deceptive_Spam_Review_Detection

Key insight: Model learns when to trust per input — combining complementary strengths

Model 2 — TextSWNN

- This model uses a two-stage CNN; it uses the concept of sentence weighting.
- Core Idea: <u>not all sentences in a review are equally important.</u>
- Some sentences (e.g., "The service was terrible") are more indicative of spam or truth than others (e.g., "We arrived at 7 PM").

Model 2 — TextSWNN

- Convert each sentence into a fixed-length vector using a convolutional neural network (CNN).
- Assign an importance weight to each sentence.
- Combine sentence vectors into a document vector using a weighted average (instead of a simple average or max pooling).
- Use that document vector for classification (spam vs. truth)

Sentence weights are computed based on word importance:

- For each word, they compute KL-divergence: a measure of how well that word separates spam from non-spam reviews.
- A sentence's weight = sum of its words' KL-divergence values, normalized across all sentences in the review.

TextSWNN · Sentence-Weighted Neural Network

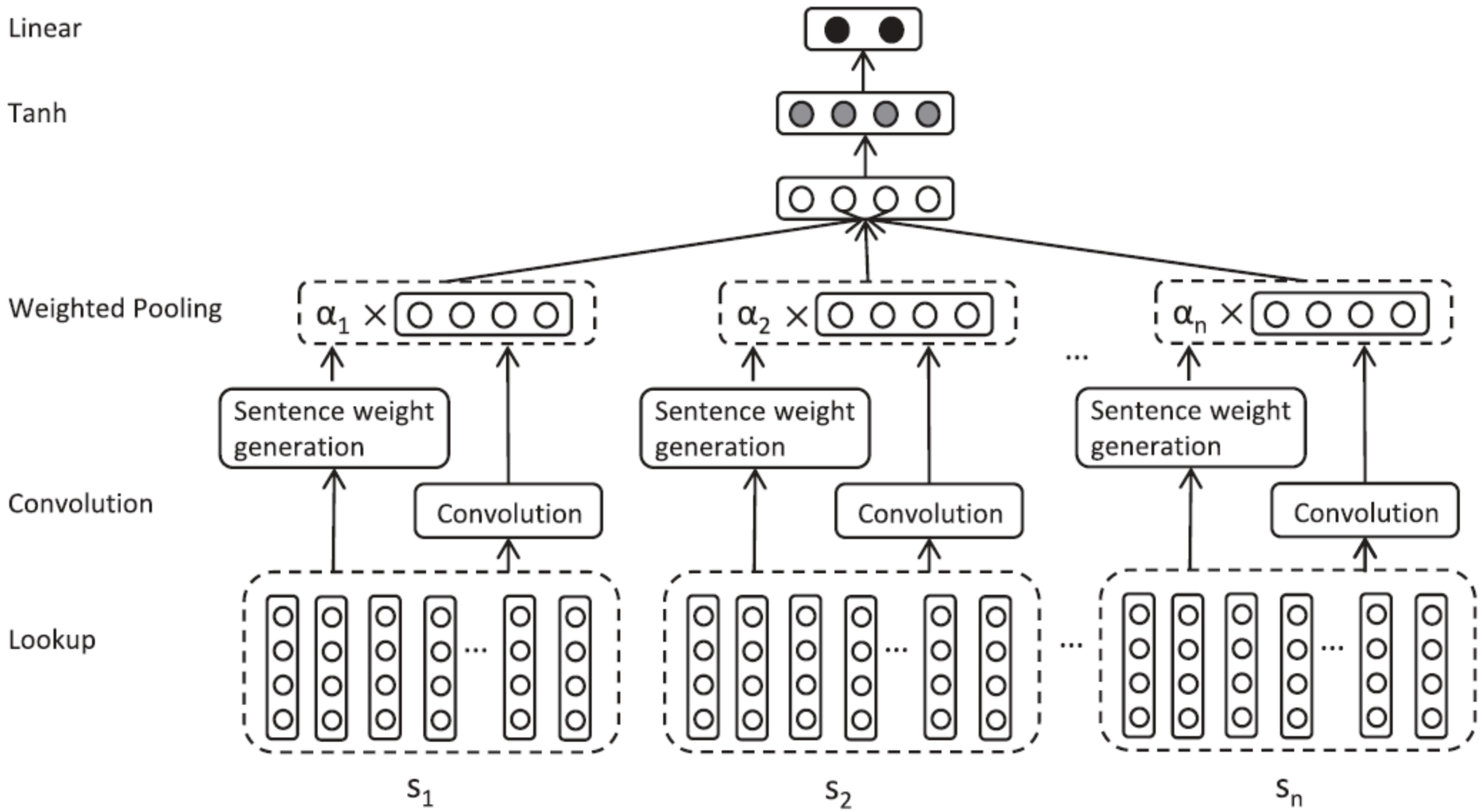


Fig. 3. SWNN model for learning sentence representation.

Changes We Introduced to the Existing CNN Architecture of the Paper

- Our TextSCNN has relearnable sentence weighting unlike the architecture of the paper (which initializes weights with KL divergence valies and never changes later). Backprop updates sentence logits, so sentence importance is relearned during training
- We used signoid function instead of tanh function at the last layer because it gave better results

Model 3 — XGBoost

- The XGBoost branch is the “structured-signal” model in the ensemble.
- It does not read raw text like RoBERTa; instead, it takes TF-IDF features from the review text plus handcrafted metadata features, then learns non-linear rules over them.
- That gives it a strong advantage on sparse, tabular-style signals that transformers do not always exploit cleanly.

Model 3 — XGBoost: Used Metadata

- Review length
  - Fake and spam reviews are often unusually short/long
- Word count
- Exclamation marks count
  - Spammy or manipulative reviews often use exaggerated punctuation to push sentiment.
- Uppercase to lowercase ratio
- Containing external URL links
- Average word length
  - very short repeated words or oddly long words can indicate low-quality generated text or unnatural phrasing (fake/spam).

Model 3 — XGBoost: Used Metadata

- Punctuation Density
  - Why it helps: spam and fake reviews often have abnormal punctuation patterns, such as too many commas, exclamation marks, or mixed punctuation.
- Type Token Ratio
  - Why it helps: this is a rough measure of lexical diversity. Repetitive, templated, or copied reviews often have lower diversity, while natural reviews usually vary their vocabulary more.
- Rating
  - Why it helps: rating is one of the strongest behavioral cues. Extremely high or low ratings that do not match the language of the review can be suspicious.

Stacking Ensemble Strategy

Level 0 — Base Models

RoBERTa +GatedFusion

TextSCNN

XGBoost +TF-IDF

3-class probabilities → 9-dim stacked feature

Level 1 — Meta-Learner

Logistic Regression Meta-Learner (trained on val-set probability stacks)

Final Prediction · 82.5% Accuracy · AUC 93.9%

Key Insight: Meta-learner learns which base model to trust per input — combining complementary strengths

Why stacking?

Each model covers a different failure mode

- RoBERTa is strong at understanding meaning and context in the review text.
- TextSCNN is good at catching local patterns like short phrases, repeated wording, and suspicious surface structure.
- XGBoost, with TF-IDF plus metadata, is good at explicit signals such as keyword frequency, review length, punctuation, rating, and category features.
- If RoBERTa is confident on a semantic case, SCNN may help on phrase-pattern cases, and XGBoost may catch obvious spam-like structure that transformer models can miss.
- The stacked model uses all three probability outputs.

Why stacking?

- Better accuracy potential
- More robustness, because mistakes from one model can be corrected by the others.
- Better generalization
- Better handling of different review styles, since fake, spam, and real reviews can fail for different reasons.

Results & Comparison (Test Set)

Model	Acc	Mac F1	AUC	Real F1	Spam F1	Fake F1	Train Time
RoBERTa+GatedFusion	80.4%	81.3%	92.4%	80.4%	75.8%	90.8%	~2374s
TextSCNN	73.4%	73.9%	87.2%	75.1%	55.7%	90.2%	~167s
XGBoost (TF-IDF+Meta)	76.7%	77.1%	90.9%	78.1%	64.5%	88.6%	~275s
Stacking Ensemble	**82.5%**	**83.5%**	**90.9%**	**82.6%**	**70.9%**	**92.0%**	**—**

REAL · F1 82.6%

Ensemble best for genuine reviews

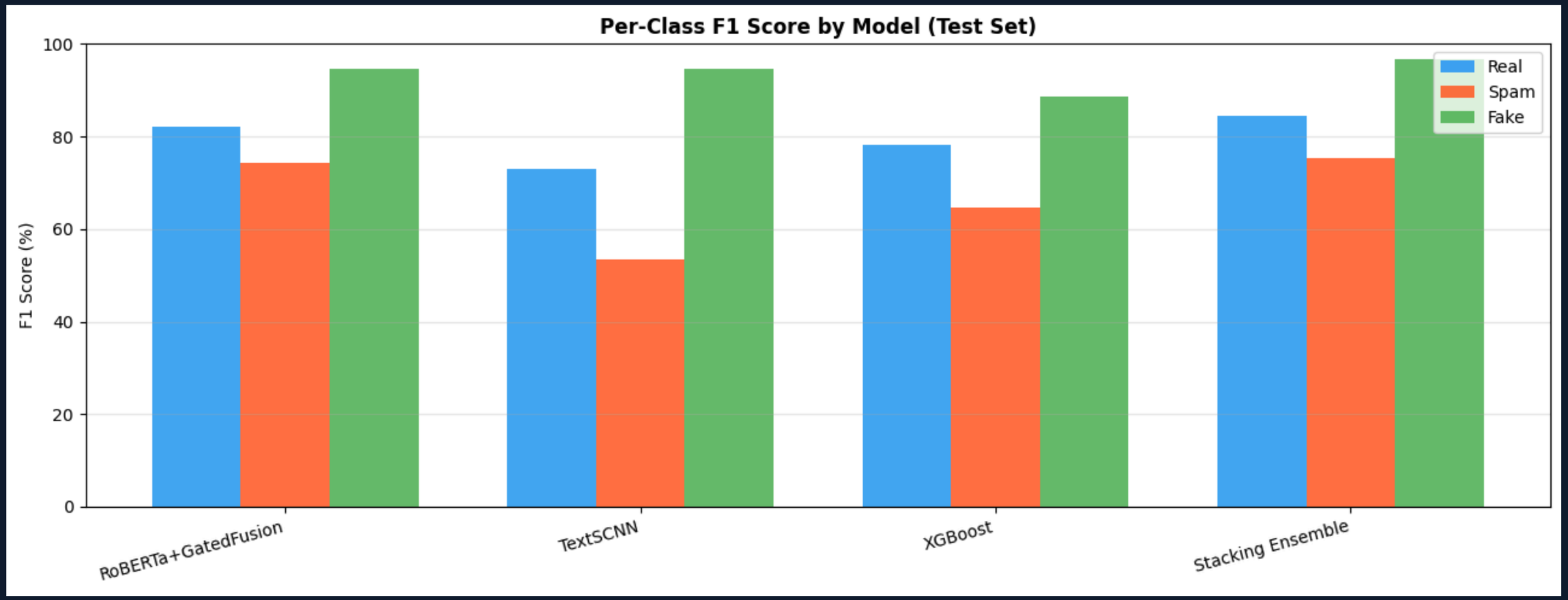
SPAM · F1 70.9%

Hardest class — gains most from ensemble

FAKE · F1 92.0%

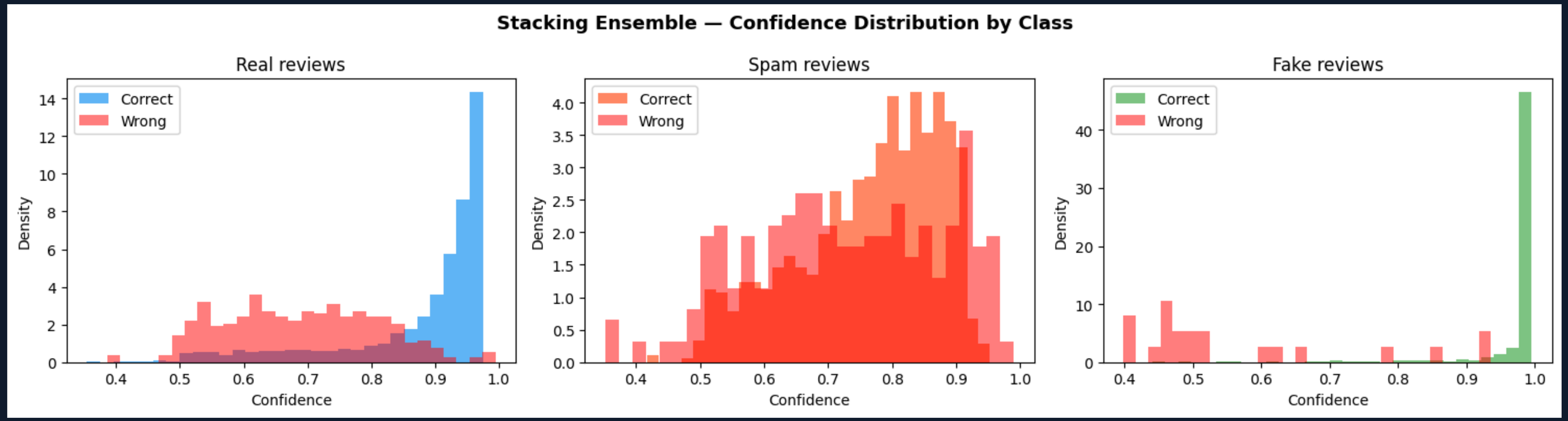
Decent detection across all models

Visual Performance Analysis



Key Takeaways: Fake class is easiest (~90-92% F1 across all models). Spam is hardest (55-71%). Ensemble consistently outperforms each base model. RoBERTa offers best accuracy/performance trade-off.

Visual Performance Analysis



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Inference & Testing System

Prediction Pipeline

1. **Input**
   Provide: review text, rating (1–5), category
2. **Extract**
   16 metadata features computed automatically
3. **Encode**
   RoBERTa tokenizer + TF–IDF vectorizer
4. **Predict**
   3 base models → probability vectors
5. **Ensemble**
   Meta-learner combines → final class label

Sample Inference Results

Good running shoes. Comfortable. Nice color. Good for running. Would recommend. Good quality. Good price.

5 ★ · Electronics

FAKE 91.3%

"Check out discounts at www.deal.com — great offers!"

4 ★ · Home

SPAM 86.8%

"Battery life is decent, screen scratches easily. Good for price."

3 ★ · Electronics

REAL 93.2%

Models saved via save_artifacts() and loadable for offline inference on any local

Metadata Analysis

SHAP (SHapley Additive exPlanations)

BERTviz Attention (Transformer Interpretability)

Review Text

Superb headphones. Crystal clear audio. Zero distortion. Perfect bass. Perfect treble. Perfect noise cancellation. Truly the pinnacle of audio engineering. Nothing comes close.

Rating

5 Stars

Category

Electronics

Analyze Review

Prediction Result

Fake

Probabilities: Real=4.4%, Spam=7.0%, Fake=88.6%

Prediction Result

Fake

Probabilities: Real=4.4%, Spam=7.0%, Fake=88.6%

Metadata SHAP (Which metadata mattered most)

avg_word_length (value: 6.4167) away
Impact: 43.11% (SHAP: -0.980151)

sentence_caps_ratio (value: 0.0452) toward
Impact: 18.27% (SHAP: 0.415325)

type_token_ratio (value: 0.875) away
Impact: 14.92% (SHAP: -0.339227)

review_length (value: 176) toward
Impact: 8.52% (SHAP: 0.193746)

category_Clothing_Shoes_and_Jewelry (value: 0) away
Impact: 8.33% (SHAP: -0.189264)

exclamation_count (value: 0) away
Impact: 2.36% (SHAP: -0.053661)

Metadata Analysis

Signals towards Fake

- Sentence caps ratio (+18.27%)
- Review length (+8.52%)

Signals away Fake

- Type-token ratio (-14.92%)
- No exclamation usage (-2.36%)

BERTviz Attention Analysis (Text-Level Importance)

High-impact tokens:

- “Perfect” (repeated) → strong repetition signal
- “audio”, “noise cancellation”, “engineering” → product terms
- Frequent “.” → short, segmented sentences

Insight: Some natural linguistic features exist, but structured positivity patterns still push prediction toward fake.

Insight: Model detects repetitive praise + fragmented sentence structure

Review Text

Bought these running shoes for half marathon training. The cushioning is great for long distances but the toe box is slightly narrow — size up if you have wide feet. Good grip on pavement and light trails.

Rating

4 Stars

Category

Sports and Outdoors

Analyze Review

Prediction Result

Real

Probabilities: Real=72.4%, Spam=27.5%, Fake=0.1%

Metadata Analysis

Signals Towards Real

- Avg. word length: +45.95% (largest contributor)
- Sentence caps ratio: +20.05%
- Type-token ratio (high lexical diversity): +12.89%
- Punctuation density: +1.35%

Signals away real

- No URL present: -11.8%
- No ALL CAPS usage: -5.52%

BERTviz Attention Analysis (Text-Level Importance)

- High-impact tokens:
  - "but" → captures contrast (pros vs cons)
  - Common structural words ("the", "is", "these") guide context
  - Punctuation (".") helps segment meaningful statements

Insight: Balanced vocabulary, moderate sentence structure, and descriptive wording strongly support authenticity.

Insight: Model focuses on contrastive reasoning (e.g., "great cushioning but narrow toe box"), a key signal of genuine reviews.

Review Text

Great blender! Also, check out my YouTube channel where I review kitchen appliances every week — link in my profile! Subscribe for more honest reviews!

Rating

5 Stars

Category

Home and Kitchen

Analyze Review

Prediction Result

Spam

Probabilities: Real=10.3%, Spam=89.7%, Fake=0.0%

Text Influence (BERT attention)

will attention score: 0.033584

I attention score: 0.03325

was attention score: 0.032861

received attention score: 0.0305

family attention score: 0.029162

I attention score: 0.028956

my attention score: 0.028762

and attention score: 0.028641

again attention score: 0.027288

this attention score: 0.026928

Higher attention score means stronger influence from the final RoBERTa attention layer.

Metadata Analysis

Signals towards spam

- Avg. word length (+34.96%)
- Presence of promotional intent (has_promo = 1)
- High lexical diversity (type-token ratio)

Signals away spam

Exclamation count (-7.94%)

BERTviz Attention Analysis (Text-Level Importance)

High-attention tokens:

- “blender”, “Great” → product + sentiment
- “link”, “Also” → transition to promotion
- “!” → emphasis / exaggeration

Insight: Features related to promotion and writing style strongly push the model toward Spam (89.7%)

Insight: Model captures shift from genuine review → self-promotion

Inference & Testing System

- SHAP and BertViz-style attention are there to make the ensemble explainable, not just predictive.
- SHAP is used for the XGBoost branch, while BertViz-style attention output is used for the RoBERTa branch.
- SHAP explains the metadata + TF-IDF part of the prediction by estimating how much each feature pushed the final decision toward Real, Spam, or Fake.

Inference & Testing System: Advantages of Using SHAP and BertViz

- SHAP gives local, per-review explanations.
- It shows which metadata features mattered most, like rating, review length, or URL presence etc.
- It helps debug whether XGBoost is over-relying on a specific feature.
- BertViz-style attention gives an intuitive word-by-word explanation.
- It is useful for spotting suspicious phrases or sentiment cues.

Thank You!!!